



EcoSmart Construction



Course Title : Computer Fundamentals

Course Code : COF101

Submitted By:

Student Name: Kaisan Arid Miah

Student ID: 242-16-026

Department of Computing and Information System (CIS)

Semester: Fall 2024

Batch: 20(A)

Submitted To:

Mr. Md. Sarwar Hossain Mollah

Associate Professor and Head, Department of **Computing and Information System(CIS)**

Daffodil International University

Table of Contents:

Introduction.....	
Task 1	
Answer to the question no.(a).....	
Answer to the question no.(b)	
Answer to the question no.(c)	
Task 2	
Answer to the question no.(a)	
Answer to the question no.(b)	
Answer to the question no.(c)	
Task 3	
Answer to the question no.(a)	
Answer to the question no.(b)	
Answer to the question no.(c)	
Conclusion	
References	

Introduction

In today's fast-paced world of technology, EcoSmart Construction understands the importance of reliable and efficient computer systems to boost productivity and ensure smooth project execution. This report outlines the key computer specifications needed for EcoSmart Construction, taking into account current market trends and budget limits. It also explores the evolution of core computer architectures, comparing older designs with modern innovations

Task1(A)

Computer Specifications and Budget Breakdown:

To meet the requirements of EcoSmart Construction, the following computer specifications have been carefully selected to balance performance and affordability:

SL	Specification	Quantity	Amount
1	Processor (Intel Core i5-14600K)	1	₺ 42,650
2	RAM (16GBDDR4)	1	₺ 7,500
3	Storage (512GB SSD)	1	₺ 3,200
4	Graphics Card (MSI GeForce RTX 3050)	1	₺ 28,800
5	Monitor (24-inch Full HD)	1	₺ 18,000
6	Operating System (Windows 11 Pro)	1	₺ 23,900
7	Peripherals (Keyboard, Mouse, UPS)	1	₺ 11,000
	Total Budget		₺ 1,35,050

Justification

To achieve an ideal balance of performance and cost, each component has been carefully chosen. The Intel Core i5-14600K processor ensures high performance, making it suitable for multitasking and complex computational tasks. Equipped with 16GB of DDR4 RAM, the system supports the smooth operation of multiple applications. 512GB SSD offers ample storage and faster data access compared to traditional hard drives. The MSI GeForce RTX 3050 is ideal for graphics-intensive applications and CAD software. 24-inch Full HD monitor provides sharp, detailed visuals, while Windows 11 pro is selected for its reliability and broad compatibility. Essential peripherals, including a keyboard, mouse, and UPS, are also included to support user interaction and data security."

Architecture of Key Computing Components:

The architecture of core computing devices has evolved considerably over time. Below is a comparison between traditional designs and modern advancements.

Sl	Feature	Older Design	Modern Design
1	CPU	Limited core count and lower clock speeds	Multiple cores, higher clock speeds (Intel Core i5-14600K)
2	Storage	Hard drive with reduced data transfer speed	High-speed SSD with improved data transfer rates
3	RAM	Low capacity (e.g., 4GB), DDR3	Enhanced capacity (e.g., 16GB), DDR4 or DDR5
4	Motherboard	Larger size and restricted connectivity options	Compact designs with extensive connectivity features (e.g., USB 3.0, USB-C)
5	Graphics Card	Standard integrated graphics	Powerful GPUs (e.g., MSI GeForce RTX 3050)
6	O Ports	Limited USB ports and outdated standards	Additional USB ports with newer standards (e.g., USB 3.0, USB-C)

Task 1 (B)

Architecture of Key Computing Devices:

Core computing devices follow the **von Neumann architecture**:

- With technological progress, the architecture of core computing devices has advanced considerably. This evolution is evident when comparing traditional designs to modern innovations. The table below highlights the key differences between older and contemporary core computing devices:



Sl	Feature	Older Design	Modern Design
----	---------	--------------	---------------

1	CPU	Single Core, Slower Clock Speeds (1-2 GHz), Higher Power Usage	Multi-Core, Higher Clock Speeds (3-5 GHz), More Energy Efficient
2	RAM	Lower Capacity (e.g., DDR2, DDR3), Higher Power Consumption	Higher Capacity (e.g., DDR4, DDR5), Lower Power Consumption
3	Storage	Hard Disk Drives (HDD), Optical Storage (CD/DVD), speeds (80GB-500GB)	Solid-State Drives (SSD), Flash Storage (e.g., USB Drives, Memory Cards), speeds (256GB-2TB)
4	Motherboard	High-speed connectivity but limited, Larger and bulkier form factors	extensive connectivity, Smaller and more compact form factors
5	Graphics Card (GPU)	Basic Integrated or Low-End discrete	High-End discrete GPUs
6	I/O Ports	Fewer USB ports, USB 2.0	More USB ports, USB 3.0/3.1, USB-C

Task 2(A)

The Role of the Operating System (OS) in Enhancing EcoSmart Construction's Performance:

The operating system (OS) functions like a manager, ensuring that all parts of a computer work together smoothly. For EcoSmart Construction, the OS plays a vital role in keeping operations running efficiently:

Resource Management: The OS controls key components like memory and the CPU, allocating resources to various programs. This ensures that multiple applications, such as project tracking tools, run simultaneously without interfering with each other—critical for EcoSmart’s data-heavy tasks.

File Organization: The OS organizes EcoSmart's essential files, such as blueprints, reports, and design documents. It makes it easy to store, retrieve, and manage these files, improving productivity and reducing time spent searching for critical information.

Multitasking: EcoSmart frequently needs several programs running at once—like project management software and live video feeds. The OS manages this multitasking effectively, preventing system freezes and allowing smooth, continuous workflow.

Security: The OS includes robust security features to protect sensitive data. It controls access to files, ensuring that only authorized users can view or modify important project details, safeguarding EcoSmart's confidential information.

By managing resources, multitasking efficiently, keeping files organized, and maintaining security, the OS helps EcoSmart Construction work more effectively, particularly on large-scale, sustainable building projects.

Task 2(B)

Why Deadlock Occurs and How to Mitigate It in Windows and Linux

Deadlock happens when programs get stuck waiting for each other, similar to a traffic jam where each car blocks the next, causing everything to stop. It occurs when multiple programs or tasks are waiting for resources held by others, leading to a freeze. This usually happens due to:

1. **Programs Holding Resources:** Programs hold resources (like memory or files) that others need, creating a bottleneck.
2. **Waiting on Each Other:** Programs can end up in a loop, where each wait for a resource the other holds, causing a deadlock.

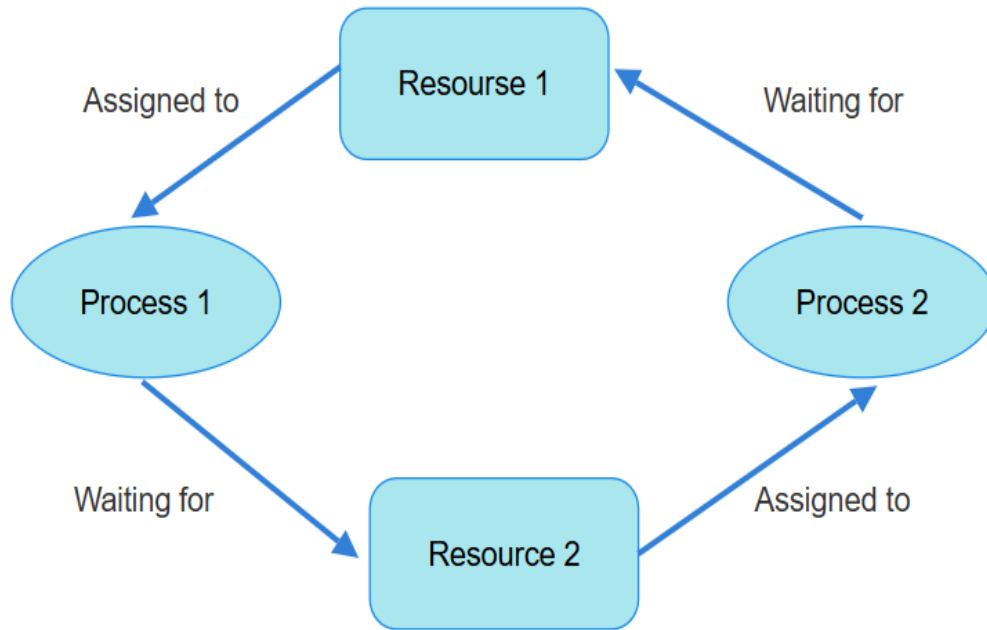


Figure: Deadlock in Operating System

Mitigating Deadlock in Windows:

- **Resource Allocation Rules:** Windows uses specific rules to allocate resources in a way that avoids deadlock, ensuring no circular waiting occurs.
- **Timeouts:** Windows may set time limits for resource requests. If a program can't get a resource in time, it releases its current resources and retries, preventing deadlock.

Mitigating Deadlock in Linux:

- **Resource Management:** Linux uses strategies like **lock ordering** to avoid circular waiting and deadlocks.
- **Deadlock Detection:** Linux detects deadlocks by checking for cycles in resource allocation. It can terminate or rollback processes to recover from a deadlock.

Here's a more concise and modified version of your explanation:

Why Deadlock Occurs and How to Mitigate It in Windows and Linux

Deadlock happens when programs get stuck waiting for each other, similar to a traffic jam where each car blocks the next, causing everything to stop. It occurs when multiple programs or tasks are waiting for resources held by others, leading to a freeze. This usually happens due to:

1. **Programs Holding Resources:** Programs hold resources (like memory or files) that others need, creating a bottleneck.
2. **Waiting on Each Other:** Programs can end up in a loop, where each wait for a resource the other holds, causing a deadlock.

Mitigating Deadlock in Windows:

- **Resource Allocation Rules:** Windows uses specific rules to allocate resources in a way that avoids deadlock, ensuring no circular waiting occurs.
- **Timeouts:** Windows may set time limits for resource requests. If a program can't get a resource in time, it releases its current resources and retries, preventing deadlock.
- **Preemption:** Windows can take resources from processes that hold them too long, preventing deadlock by ensuring no process monopolizes resources.

Mitigating Deadlock in Linux:

- **Resource Management:** Linux uses strategies like **lock ordering** to avoid circular waiting and deadlocks.
- **Deadlock Detection:** Linux detects deadlocks by checking for cycles in resource allocation. It can terminate or rollback processes to recover from a deadlock.
- **Timeouts:** Linux uses timeouts and retries to prevent processes from waiting indefinitely.

General Deadlock Prevention:

- **Deadlock Prevention:** Both systems can prevent deadlock by ensuring resources are requested in a specific order and using timeouts.
- **Deadlock Detection and Recovery:** Both Windows and Linux detect deadlocks and recover by killing or rolling back processes involved.

Task 3(A)

The EcoSmart Construction network diagram shows the company's internal network layout, ensuring efficient communication, secure data management, and optimal resource use.

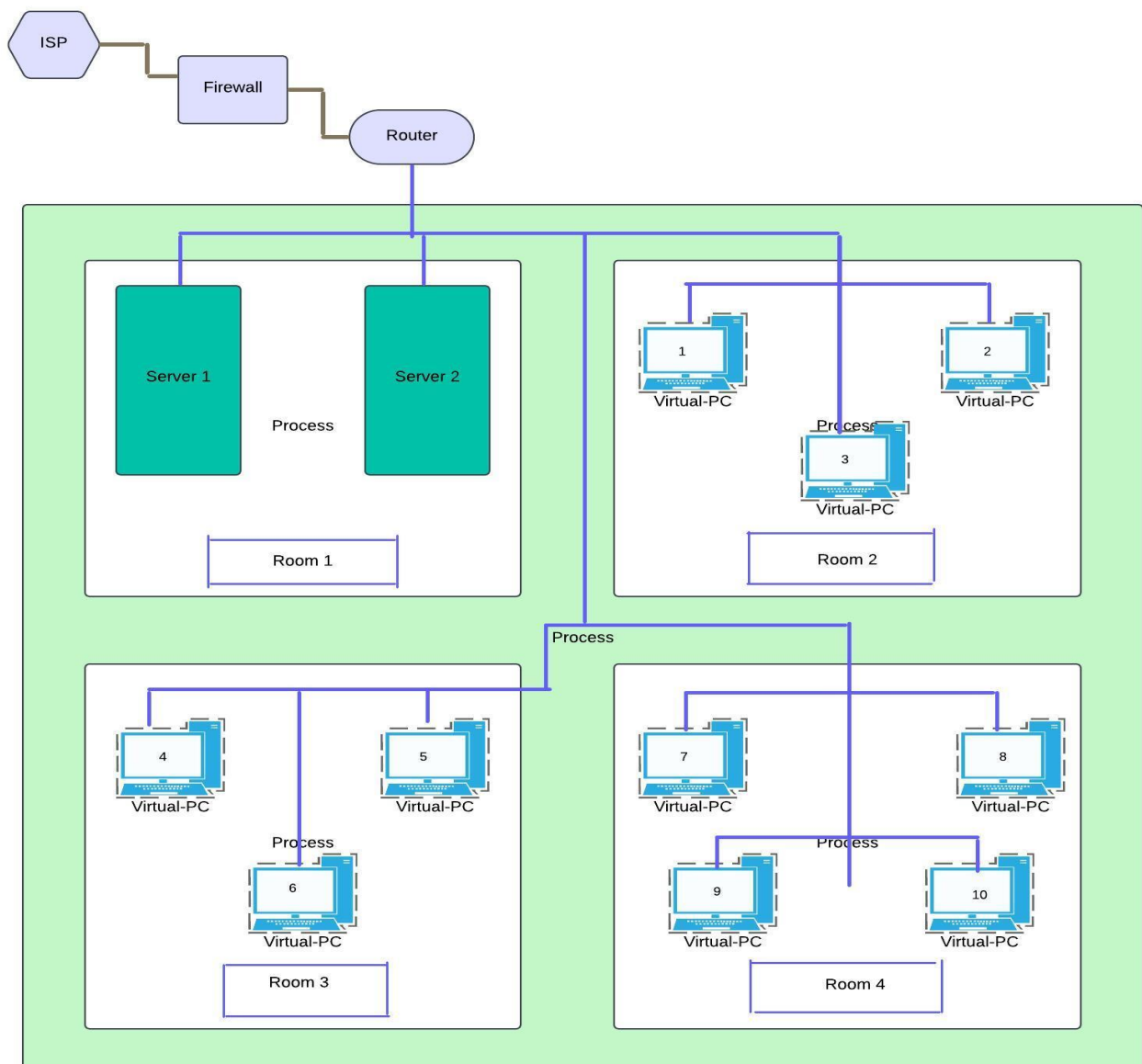


Figure : Computer Network Diagram for EcoSmart Construction with Specific Roles

Network Components and Their Roles change the line

Network components are essential parts of any network infrastructure, and each component has a specific role in ensuring smooth communication and data transfer. Here's a brief overview of common network components and their functions:

Firewall: Protects networks from unauthorized access by monitoring and controlling incoming and outgoing traffic. It can be either hardware or software-based.

Router: Directs data packets between networks, typically between a local network and the internet. It helps manage traffic and determine the best path for data.

An **Internet Service Provider (ISP)** is a company that provides internet access to users. They offer various connection types like broadband or fiber, and services such as email, web hosting, and IP addresses. ISPs ensure stable internet connectivity and may also offer extra services like security software. Examples include AT&T and Grameenphone.

The **File Server** (IP **192.168.10.1.2**) stores and manages company files, making them easily accessible and shareable among employees. It is connected directly to the router, enabling file access throughout the entire network. This centralized storage system helps in efficient document management across the organization.

The **Backup Server** (IP: 192.168.10.11) ensures the regular backup of critical data, allowing recovery in case of system failures or data loss. It is connected to the router and regularly syncs with the file server to maintain up-to-date backups, providing a reliable data protection solution across the network.

Room 1 Switch: connects devices within the room to the network, allowing communication with servers.

Design PCs 1 and 2: are workstations used by the design team for project creation and editing. Both PCs rely on the switch for network access.

Room 2 Switch: connects devices within the room to the network, allowing access to servers.

Finance PCs 1 and 2: are used by the finance team for managing financial tasks such as budgeting and accounting, with the switch providing the necessary network access.

The network diagram effectively shows how key components like the **ISP, firewall, router, servers**, and devices in **Room 1** and **Room 2** are connected. Each device's role and its associated IP address are clearly outlined, ensuring a structured and efficient network. This setup facilitates secure communication, streamlined data management, and smooth operations at EcoSmart Construction. The integration of these elements supports the company's IT infrastructure, ensuring both security and operational effectiveness.

Task 3(B)

At **EcoSmart Construction**, ensuring the protection of both physical and digital assets is critical to maintaining the security and integrity of operations. This policy provides key guidelines for employee behavior and outlines the security measures required to safeguard valuable resources. These include proper handling of sensitive information, adherence to access controls, and protocols for securing data both digitally and physically. The goal is to maintain a secure environment where business continuity and employee safety are prioritized, helping to prevent data breaches, theft, or any compromise of organizational assets.

Revised Security Policy for EcoSmart Construction

Employee Conduct:

The employee conduct policy emphasizes the importance of confidentiality, secure access control, and proactive measures to protect company data. Employees are expected to manage their passwords securely, lock workstations when not in use, avoid risky internet behaviors, and exercise caution with emails. Prompt reporting of any security issues is essential to maintain the integrity of the organization's operations and safeguard against potential threats. This policy promotes a culture of security awareness and responsibility within the company.

Physical Security Measures:

Physical security at EcoSmart Construction involves strict access control through cards or biometrics, careful management of visitors, surveillance through CCTV cameras, and regular patrols by security personnel. Additionally, all assets are inventoried and securely stored to

protect against theft or loss. These measures are designed to safeguard the company's physical infrastructure and assets effectively.

Digital Security Measures:

Digital security at EcoSmart Construction involves multiple layers of protection, such as data encryption, firewalls, antivirus software, and regular backups. Role-based access control helps secure sensitive data, while patch management keeps systems secure from vulnerabilities. Monitoring network traffic and providing regular training for employees ensures the company is protected from both internal and external cyber threats. These measures are crucial for maintaining the integrity and confidentiality of the company's digital assets.

Incident Response Plan:

An effective incident response plan at EcoSmart Construction ensures quick identification and containment of security incidents, followed by threat removal and system recovery from backups. After resolving the issue, a review is conducted to prevent future incidents, and all actions are properly documented and reported for transparency and accountability.

Compliance and Enforcement:

At EcoSmart Construction, compliance with security policies is mandatory for all employees, with strict consequences for non-compliance. Regular audits are conducted to ensure the policies are followed, which helps maintain a secure operational environment and protects company resources.

By following these guidelines, EcoSmart Construction ensures the security and resilience of its operations. Adhering to the policy is critical for protecting assets and maintaining the integrity of daily operations. The company commits to reviewing and updating security measures regularly to address emerging threats. A strong culture of security and vigilance will help safeguard both business and resources, ensuring long-term success.

Conclusion

In conclusion, this proposal for EcoSmart Construction underscores the importance of integrating modern technology solutions and network designs with the company's operational objectives. By thoughtfully incorporating high-performance hardware, dependable software, and a robust network infrastructure, EcoSmart is positioned to boost efficiency, streamline project workflows, and inspire innovation. The implementation of cutting-edge security protocols and effective data management strategies ensures the protection of vital assets and smooth operational processes. This comprehensive approach not only addresses current needs but also sets up a flexible framework to support future growth and advancements in technology. EcoSmart Construction's focus on utilizing technology demonstrates its commitment to leading in sustainable and efficient construction practices

Reference:

1. https://www.amazon.com/s?k=Intel+Core+i5-14600K+price&crid=2D6V6J2N987U&sprefix=intel+core+i5-14600k+price%2Caps%2C321&ref=nb_sb_noss
2. https://www.amazon.com/s?k=MSI+GeForce+RTX+3050+buy&crid=RNF8XNYDS39&sprefix=msi+geforce+rtx+3050+buy%2Caps%2C903&ref=nb_sb_noss
3. <https://www.microsoft.com/en-us/>
4. Geeks for Geeks - Operating System: <https://www.geeksforgeeks.org/operating-system>
5. Tutorials Point - OS Deadlocks: https://www.tutorialspoint.com/operating_system/os_deadlocks.htm
6. Tutorials Point: https://www.tutorialspoint.com/operating_system/os_deadlocks
7. Kaspersky Cybersecurity: <https://www.kaspersky.com/resource-center>